Vision2Text: Evaluating Image-to-Text Generation Models for Semantic Accuracy and Hallucination

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Abstract

Recent advances in vision-language models have improved machines' ability to generate natural language descriptions of images. Although modern captioning systems produce fluent and semantically rich captions, they often suffer from object hallucinations where descriptions include objects not present in the image. These errors reduce the reliability of captioning systems in real-world applications. In this work, we evaluate several pretrained image captioning models and analyze their hallucination behavior using a unified evaluation setup. To better measure factual grounding, we introduce a novel metric, the Grounded Semantic Consistency Score (GSCS). It compares objects mentioned in generated captions with those detected in the corresponding image. This metric provides a direct measure of semantic grounding and helps distinguish linguistic fluency from factual correctness in image captioning models.

1. Introduction

Automatically describing visual scenes in natural language is a key challenge in multimodal artificial intelligence. Image captioning systems connect visual understanding and language generation by producing sentences that summarize an image's content. This capability has important applications in assistive technologies, image retrieval, and human-AI interaction.

Recent progress in transformer-based vision language models has greatly improved caption generation. Modern architectures produce fluent, contextually meaningful descriptions using large pretrained models without task-specific training. However, despite strong performance on standard metrics, these models often generate captions with objects or attributes not present in the image. This phenomenon, called *object hallucination*, occurs when language priors outweigh visual grounding during generation. The metrics used make this hard to detect: BLEU, METEOR, and CIDEr compare captions only to human references and ignore the image, while CLIPScore considers the image but rewards overall scene plausibility rather than per-object accuracy.

In this work, we analyze the hallucination behavior of several pretrained image captioning models. To better evaluate semantic grounding, we introduce a new metric, the *Grounded Semantic Consistency Score (GSCS)*, which measures consistency between objects mentioned in generated captions and those detected in the image. By combining caption generation with visual object detection, our approach offers a more direct assessment of factual grounding than standard metrics. Unlike prior hallucination metrics, it uses the image itself, not reference captions, as the source of truth.

2. Related Work

2.1. Image Captioning Models

A popular early model, "Show and Tell" [14], used a CNN encoder and an RNN decoder to produce captions. It converted the image into a feature vector fed into an LSTM that sampled the most probable word for the sentence one at a time. This was the first paper to combine computer vision and machine translation in a single encoder-decoder framework. Later work followed the same architecture but strengthened the visual encoder. For example, by swapping a GoogLeNet CNN for a pretrained ResNet50 with soft attention, Chu et al. achieved significant gains [3].

Following the popularization of transformer and attention architectures, models such as ClipCap used the CLIP encoder and the GPT-2 decoder for captioning [8]. Similar models like ViT-GPT2 use a Vision Transformer instead of CLIP and retain the GPT-2 decoder. They proved more powerful than previous CNN-RNN-style architectures [5]. Currently, a competitive state-of-the-art style model is BLIP (Bootstrapping Language Image Pretraining) [6]. BLIP consists of three transformer components: a ViT, a Text Transformer, and a Cross-Modal Transformer that fuses text and visual features. BLIP's single multimodal approach is flexible, modular, and powerful.

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While early CNN-RNN models focused on feature vectors and sequential decoding, transformer-based image captioners unify encoding and decoding with attention. We evaluate these three families (BLIP, ViT-GPT2, and ClipCap). Our contribution is a new way of measuring how often they hallucinate, not a new captioner.

2.2. Hallucination in Image Captioning

These models are not perfect, partly due to object hallucination, where a model generates captions including objects not present in the image. Despite performance improvements, contemporary image captioning models still hallucinate objects [12]. Rohrbach et al. devised CHAIR (Caption Hallucination Assessment with Image Relevance), a metric measuring object hallucination by calculating the proportion of generated words corresponding to objects actually present in the image. They found that on the Karpathy test set, between 7.4 and 17.7 percent of sentences contain hallucinated objects despite competitive performance on standard metrics [12].

While CHAIR was innovative, it depended on fixed MS COCO objects and synonyms. Aloha (Assessment with Language models for Object Hallucination) uses a large language model to enable open-vocabulary hallucination detection [10]. It allows flexible scoring and handles uncertainty. Both CHAIR and ALOHa address the right problem but keep the *reference captions* as the source of truth, penalizing models whose vocabulary diverges from the references even when captions are factually correct. Our metric (§3) instead uses what is actually in the image, confirmed by an object detector, as the source of truth.

2.3. Object Detection for Visual Grounding

Visual grounding essentially performs the opposite of image captioning: it takes a description and tries to find the region of the image that corresponds to it. This is where models like Grounding DINO really excel [7]. Grounding DINO marries DINO (a transformer-based detector) with grounded pretraining. It addresses the need for a robust detector in open-world scenarios, including zero-shot transfer, expression comprehension, and object detection. Its main strength lies in its open vocabulary, which allows it to detect objects not encountered during training. It is precisely this open vocabulary that lets us use it as the evidence source for our metric.

3. Method

3.1. Dataset

The evaluation dataset is Flickr30k [11] because our models are not trained on it. Flickr30k contains 30,000 diverse real-world images that are human-annotated and widely used as a benchmark for image captioning. We evaluate only the first 50 samples (sorted by filename) using `SEED=42`. This ensures all three models are compared on the same images with the same detector, extractor, and metric settings. The small sample caveat is discussed in §4.4.

3.2. Captioning Models

ClipCap pairs a frozen CLIP visual encoder with a GPT-2 decoder via a light mapping network finetuned for caption generation [8]. As the lightest model we evaluate, we treat it as the baseline. **ViT-GPT2** pairs a Vision Transformer encoder with a GPT-2 language decoder [5] and serves as a strong supervised baseline. **BLIP** [6] is a unified vision-language architecture with multimodal pretraining. It is the strongest model in our experiments and the state of the art. All three are evaluated zero-shot on Flickr30k; specific checkpoints and training data are listed in Appendix A.2.

3.3. Evaluation Metrics

BLEU [9] measures n-gram precision against the references. **METEOR** [1] adds stemming and synonym matching for better recall. **CIDEr** [13] weights n-gram consensus by TF-IDF. All three consider only words and not the image. Thus, a caption can include nonexistent objects yet still score well if enough common phrases overlap with a reference. Their closed forms are reproduced in Appendix A.1.

CLIPScore is reference free and image aware [4]:

$$\text{CLIPScore} = \max\big(0, \cos\big(f_I(I), f_T(T)\big)\big)\,, \quad (1)$$

where f_I and f_T are CLIP's image and text encoders. CLIPScore captures overall scene alignment but its coarse granularity may fail to penalize fine-grained hallucinations, such as small nonexistent objects, as long as the overall semantics remain consistent.

3.3.1. Grounded Semantic Consistency Score

Similar to CHAIR or ALOHa, we believe a high BLEU or CLIPScore does not guarantee factual grounding [10, 12]. We propose a novel metric, the Grounded Semantic Consistency Score (GSCS), to evaluate factual grounding of a caption. GSCS extracts objects from generated captions using NLTK [2], detects visual objects in the corresponding image with Grounding DINO [7], and reports the grounding precision:

$$\text{GSCS} = \frac{TP}{TP + FP} = \frac{|\mathcal{O}_{\text{cap}} \cap \mathcal{O}_{\text{det}}|}{|\mathcal{O}_{\text{cap}}|}, \quad (2)$$

where $\mathcal{O}_{\text{cap}}$ is the set of objects extracted from the caption and $\mathcal{O}_{\text{det}}$ is the set of objects detected in the image. GSCS avoids the ground truth caption dependency of CHAIR [12] and the semantic parsing of ALOHa [10], relying solely on visual evidence. A high GSCS indicates strong factual grounding and a low GSCS indicates hallucination. Unlike traditional metrics, GSCS explicitly penalizes unsupported object mentions and separates semantic reliability from linguistic fluency. By design, GSCS is a precision-style score: it penalizes captions for hallucinations but not omissions. We chose this asymmetry because recall against the detector would punish concise captions rather than wrong ones and because Grounding DINO is imperfect at recall.

3.4. Implementation

We use Grounding DINO at box $\tau = 0.35$ and text $\tau = 0.25$, with NLTK as the noun extractor and the WordNet lemmatizer. We filter eight meta-words (`image`, `photo`, `picture`, `scene`, `background`, `front`, `top`, `side`) from $\mathcal{O}_{\text{cap}}$ before intersecting with $\mathcal{O}_{\text{det}}$ since no object detector can ground them. §4 ablates this choice. If a caption has zero extracted nouns, GSCS is reported as 1 and the row is included in the average. Captions are decoded with beam search (beams = 4, max_length = 24). Full hyperparameters and checkpoint names are listed in Appendix A.2. The full implementation can be found at `https://github.com/wz-cutting-edge/Vision2Text`.

4. Experiments and Results

4.1. Results

Model	BLEU	METEOR	CIDEr	CLIPScore	GSCS
BLIP	**0.202**	0.330	0.436	**27.70**	**0.843**
ViT-GPT2	0.177	**0.349**	**0.465**	27.56	0.706
ClipCap	0.000	0.188	0.011	20.39	0.742

Table 1. Evaluation results across the three captioners on the same 50-image Flickr30k subset, scored with the metrics in §3.3. Bold marks the best per column. ClipCap's BLEU collapses to 0 because its short captions rarely share a 4-gram with a reference.

Although BLIP and ViT-GPT2 have similar CLIPScores, BLIP performs noticeably better on GSCS (Table 1), grounding itself better than ViT-GPT2 even though CLIP cannot distinguish them. This is likely because ViT-GPT2 outputs longer sentences with more ambitious captioning (averaging 3 nouns per caption versus BLIP's 2.4), increasing its chances to hallucinate. ClipCap's BLEU collapses to about 0, with low overall scores, yet it beats ViT-GPT2 on GSCS. One explanation is that ClipCap's captions are not very fluent or semantically correct but tend to predict objects well, while ViT-GPT2's error rate grows with the number of objects it predicts. We caution that the ClipCap vs. ViT-GPT2 ranking on GSCS partly reflects the meta-word filter, as discussed in §4.2.3. Overall, BLIP stands out across all metrics, consistently scoring highest or second highest by a close margin, which makes sense because BLIP refines its captions synthetically. BLIP's lead is widest on GSCS and narrowest on CLIPScore, suggesting that CLIP's coarse alignment misses the per-object errors GSCS is designed to catch (Figure 3).

4.2. Ablation Experiments

We run three ablations on GSCS: a Grounding DINO threshold sweep, a POS tagger swap, and an experiment on the effect of the `IGNORE_WORDS` filter. Together, they let us see how the components of Eq. 2 behave.

4.2.1. Grounding DINO Thresholds

Variant	box τ	text τ	BLIP	ViT-GPT2	ClipCap
Strict	0.45	0.35	**0.613**	0.512	0.435
Tight* (def.)	0.35	0.25	**0.843**	0.706	0.742
Loose	0.25	0.15	**0.848**	0.797	0.822
Permissive	0.15	0.10	0.900	0.851	**0.904**

Table 2. GSCS across four Grounding DINO threshold variants; * marks the paper default used in Table 1. Bold marks the best model per row. Higher τ requires the detector to be more confident; lower τ does the opposite. Appendix A.3 expands on box τ vs. text τ.

We sweep four settings: strict, tight, loose, and permissive (Table 2). BLIP comes out as the strongest performer overall, winning three of four settings and tying with ClipCap on permissive. ClipCap is the most threshold sensitive: its captions rely on broad locations such as `park`, `street`, and `restaurant`, which Grounding DINO accepts easily at low thresholds but not at box τ=0.45. The takeaway is that GSCS is a good *directional* signal (BLIP wins almost everywhere). Still, the absolute number depends on the threshold, so we recommend reporting alongside an alternative setting. Figure 5 visualizes the same sweep as a line chart, making the monotonic rise and the BLIP vs. ClipCap crossover near Permissive immediate.

4.2.2. POS Tagger

We swap the NLTK POS tagger with spaCy and find that GSCS changes by at most 0.026 across all three models, so our reported numbers are robust to the choice of noun extractor. A deliberately naive extractor (regex tokens, no POS tagging) halves GSCS for every model, establishing that POS tagging is necessary, but any specific toolkit will do. The full per model table is in Appendix A.4.

4.2.3. Ignore Words List

Model	Filter on (default)	Filter off	Δ
BLIP	0.843	0.817	+0.027
ViT-GPT2	0.706	0.676	+0.030
ClipCap	0.742	0.575	+0.167

Table 3. Effect of the eight word `IGNORE_WORDS` filter on GSCS. "Filter on" is the paper default in Table 1; "filter off" includes the meta-words `image`, `photo`, `picture`, `scene`, `background`, `front`, `top`, `side` in $\mathcal{O}_{\text{cap}}$. ClipCap's Δ is six times larger than the others'.

IGNORE_WORDS removes meta-words from $\mathcal{O}_{\text{cap}}$ before intersecting with $\mathcal{O}_{\text{det}}$ because the object detector cannot ground them. Table 3 shows that BLIP and ViT-GPT2 are largely unaffected, while ClipCap shows a large jump. This likely reflects the specific ClipCap checkpoint we evaluated, which is finetuned to generate high-level narrations that include many meta-words. With the filter off, ClipCap drops below ViT-GPT2 (0.575 vs. 0.676), so the ClipCap vs. ViT-GPT2 ranking from Table 1 depends on the filter. We conclude IGNORE_WORDS is necessary, since otherwise GSCS penalizes captions for words the detector cannot ground, rather than genuine hallucinations. Any reported GSCS number should disclose whether this filter is applied.

4.3. Hallucination Comparisons

Model	GSCS	1-CHAIR_i	CHAIR_s↓	ALOHa	ALOHa_s↓
BLIP	**0.843**	**0.645**	**0.60**	**0.821**	**0.30**
ViT-GPT2	0.706	0.556	0.74	0.771	0.58
ClipCap	0.742	0.150	0.86	0.524	0.74

Table 4. Hallucination and grounding metrics on the same 50-image subset. Higher is better for GSCS, $1-\text{CHAIR}_i$, and ALOHa; lower is better for the columns marked ↓. Bold marks the best per column. BLIP wins under every metric. GSCS is the only metric under which ClipCap ranks above ViT-GPT2.

To assess whether GSCS adds information rather than merely duplicating other hallucination metrics, we compare it with CHAIR_i, CHAIR_s, ALOHa, and ALOHa_s. Table 4 (and the bar view in Figure 4) shows all four metrics agree that BLIP is the strongest captioner, removing concern that BLIP's top ranking is specific to GSCS. GSCS is the only metric where ClipCap ranks above ViT-GPT2: ClipCap's captions rely on broad scene labels that Grounding DINO confirms easily, but rarely appear in human references, so reference-based metrics penalize them. GSCS measures object presence, while CHAIR and ALOHa measure object vocabulary overlap with references; when a model's vocabulary diverges, the results differ. Per-image Spearman correlations between GSCS and CHAIR_i / ALOHa are weak across all models ($\rho \in [-0.18, +0.37]$); per-model heatmaps in Figure 1 show this independence visually, confirming GSCS adds a signal at the per-caption level rather than rediscovering CHAIR or ALOHa. Practitioners choosing between BLIP and ClipCap on reference-based metrics get a different signal than on GSCS. That disagreement is informative: it points to a vocabulary mismatch with references rather than outright fabrication. We recommend reporting GSCS alongside CHAIR or ALOHa, not instead of them.

4.4. Limitations

The main limitations stem from limited computing power. We evaluate only 50 of the 30,000 images in Flickr30k, a small sample size enough to provide preliminary evidence that GSCS behaves sensibly but not to draw tight conclusions. Differences smaller than about 0.05 in GSCS should not be overinterpreted, and the ClipCap vs. ViT-GPT2 swap (0.742 vs. 0.706) falls within that margin. The detector is also a limitation: we use Grounding-DINO-tiny, which can penalize GSCS when it misses less obvious objects; a larger detector like Grounding DINO Large would help. Grounding DINO detects only nouns, not verbs, so hallucinated actions go uncaught. For example, a caption may say a man is running when he is standing; as long as the man is detected, GSCS scores it as correct. Similarly, GSCS checks object presence, not semantic correctness: "a man is riding a dog" would score 1.0 even though the relation is absurd. Lastly, threshold sensitivity is real (Table 2), so we recommend reporting GSCS at an alternative threshold alongside the default.

5. Conclusion

This paper proposes the Grounded Semantic Consistency Score (GSCS), defined in Eq. 2, to evaluate hallucinations in pretrained image captioning models in zero-shot scenarios. Unlike reference-based metrics such as BLEU, METEOR, and CIDEr, GSCS uses an open-vocabulary object detector as the source of truth. This lets it penalize fabricated content that previous metrics overlook. Our experiments confirm that GSCS distinguishes models with similar CLIPScore (Table 1), is consistent across detector thresholds (Table 2), and adds an independent signal beyond CHAIR and ALOHa (Table 4). The runner-up ranking depends on the meta-word filter (§4.2.3). This is the kind of design choice we want a well-behaved metric to expose. Future work includes scaling to more images and integrating relation-level grounding to extend GSCS to verbs and attributes, not just nouns.

6. Peer Evaluation

Aluri Sri Moukthika 20	Myself 20

Table 5. Self-Peer Evaluation Table

Work was divided as follows: William contributed to ClipCap and BLIP, conducted ablation studies, and performed hallucination metric comparisons, while Sri implemented ViT-GPT2, completed ClipCap, and contributed to BLIP.

For the presentation, Sri covered the motivation, objectives, portions of the models, dataset, metrics, examples, and conclusion. William handled model comparisons, evaluation metrics, the proposed metric, ablation studies, results, limitations, and examples.

For the paper, Sri authored the Abstract, Introduction, and parts of Related Work and Methods, while William contributed to Related Work, Methods, Experiments and Results, Conclusion, and Appendix.

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A. Appendix

A.1. Reference Metric Equations

For completeness, the closed forms of the three reference-based metrics in §3.3 follow:

$$\mathrm{BLEU} = \mathrm{BP} \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right), \tag{3}$$

where p_n is the modified n-gram precision, $w_n = 1/N$ is the uniform weight, and the brevity penalty is

$$\mathrm{BP} = \begin{cases} 1 & \text{if } c > r, \\ e^{(1-r/c)} & \text{if } c \leq r, \end{cases} \tag{4}$$

with c the candidate length and r the reference length [9].

$$\mathrm{METEOR} = F_{\text{mean}} \cdot (1 - \mathrm{Penalty}), \tag{5}$$

where F_{mean} is a recall biased harmonic mean of unigram precision and recall, and $\mathrm{Penalty}$ punishes word order mismatches [1].

$$\mathrm{CIDEr} = \frac{1}{N} \sum_{n=1}^{N} \frac{g_c^n \cdot g_r^n}{\|g_c^n\| \, \|g_r^n\|}, \tag{6}$$

where g_c^n and g_r^n are TF-IDF weighted n-gram vectors for the candidate and reference [13].

A.2. Implementation Details

We split the models into three notebooks, each using the first 50 images sorted by filename, with `SEED=42` set via `numpy.random.seed` and `torch.manual_seed`. Per-model checkpoints, training data, and decoding setup appear in Table 6.

Model	Checkpoint
BLIP	`Salesforce/blip-image-captioning-base`
ViT-GPT2	`nlpconnect/vit-gpt2-image-captioning`
ClipCap	`michelecafagna26/clipcap-base-captioning-ft-hl-narratives`

Table 6. Pretrained checkpoints used for each captioning model.

All models are initialized from HuggingFace checkpoints and evaluated zero-shot on Flickr30k. BLIP and ViT-GPT2 are trained on the MS COCO dataset, while ClipCap is fine-tuned on HL Narratives based on an MS COCO base. For decoding, BLIP and ViT-GPT2 use beam search with a beam size of 4, a maximum generation length of 24 tokens, and early stopping. ClipCap uses the `generate_beam` implementation from the original repository.

The object detector is `IDEA-Research/grounding-dino-tiny` [7], used zero-shot with the per-noun query template `"a {w}"` at box $\tau = 0.35$ and text $\tau = 0.25$. The noun extractor uses NLTK [2] with `word_tokenize`, `averaged_perceptron_tagger_eng`, and the WordNet lemmatizer (POS $\in$ {NN, NNS, NNP, NNPS}, alphabetic lemmas only). CLIPScore uses OpenAI CLIP ViT-B/32 scaled by $100 \cdot \max(0, \cos(I, T))$. CIDEr uses `pycocoevalcap.cider.Cider`, and BLEU and METEOR use HuggingFace `evaluate.load("bleu")` and `evaluate.load("meteor")` with default settings.

A.3. Box and Text Thresholds in Grounding DINO

For Table 2: box τ is the confidence threshold on predicted bounding boxes. A higher box τ means fewer boxes with higher precision and lower recall. A lower box τ gives more boxes with higher recall and lower precision. text τ is the similarity threshold between an image region and the text prompt. A higher text τ means stricter grounding to the prompt and may miss valid but less obvious matches. A lower text τ means looser semantic matching with a risk of *semantic drift*, where the wrong object is matched to the text [7].

A.4. POS-Tagger Ablation

Holding thresholds at the paper default $(\text{box}\,\tau, \text{text}\,\tau) = (0.35, 0.25)$, we swap the noun extractor among NLTK [2], spaCy, and a deliberately naive regex plus stopword baseline.

Model	**NLTK***	**spaCy**	**Naive**
BLIP	0.843	0.837	0.453
ViT-GPT2	0.706	0.709	0.330
ClipCap	0.742	0.716	0.306

Table 7. GSCS under three noun extractors; * marks the paper default. NLTK and spaCy agree to within 0.026 across all models. The naive extractor pulls non-noun tokens into $\mathcal{O}_{\text{cap}}$ that the detector cannot ground, halving GSCS and confirming that POS tagging, but not any specific toolkit, is necessary.

A.5. Vocabulary-Level Failure Modes

A look at which nouns each model gets right vs. wrong is revealing. **BLIP**'s top hallucinated nouns (`man`, `street`, `group`) are also its top reliably grounded nouns; its errors are common, visually plausible nouns rather than exotic ones. **ViT-GPT2** produces `cell phone`, `parking lot`, and `snow` on Flickr30k images that contain none of them. These are high-frequency MS COCO 2017 categories, a textbook example of training distribution priors leaking into generation. **ClipCap**'s top hallucinations are abstract or activity nouns (`fun`, `living`, `game`, `tennis`) that an object detector cannot, in principle, attempt to ground. This is the failure mode that our discussion of limitations in §4.4 refers to, and Figure 2 renders the ViT-GPT2 case as a side-by-side bar chart of hallucinated versus reliably grounded nouns.

A.6. Figures

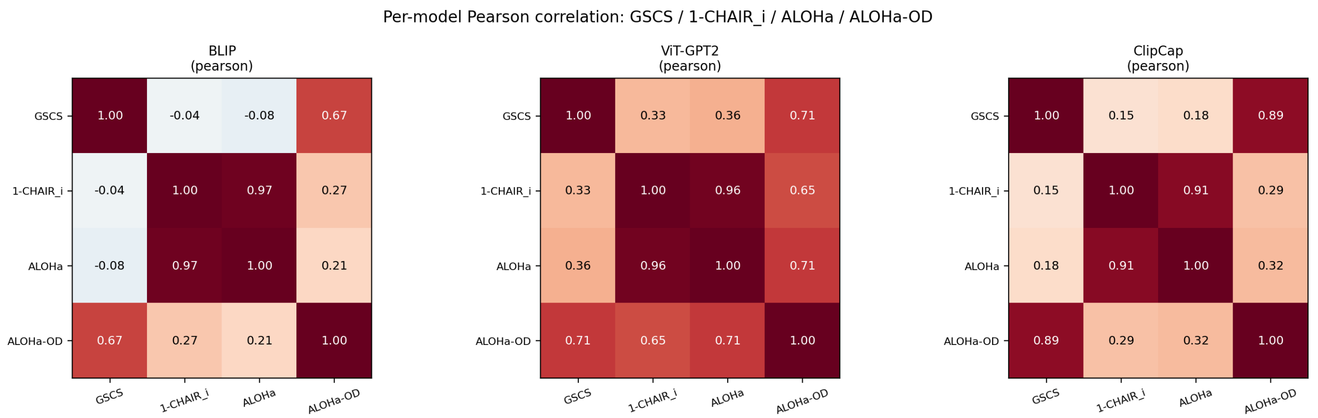

Figure 1. Per-model 4x4 Pearson correlation matrices over (GSCS, 1−CHAIR_i, ALOHa, ALOHa-OD). The (GSCS, CHAIR) and (GSCS, ALOHa) cells are consistently the lightest, while (CHAIR, ALOHa) is dark, confirming that GSCS measures an independent signal at the per-image level.

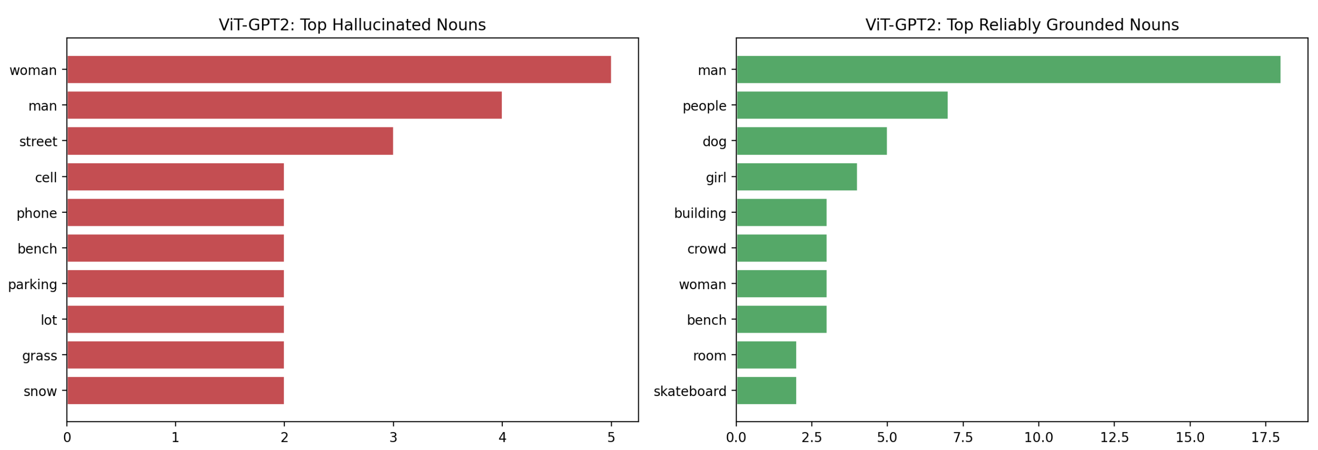

Figure 2. Top-10 hallucinated nouns (left, red) versus top-10 reliably grounded nouns (right, green) for ViT-GPT2 on the 50-image Flickr30k subset. A noun is counted as hallucinated when ViT-GPT2 mentions it, but Grounding DINO does not detect it in the corresponding image. Tokens such as `cell phone`, `parking lot`, and `snow` are MS COCO 2017 categories absent from these Flickr30k images, providing a concrete example of training-distribution priors leaking into caption generation, as discussed in Appendix A.5.

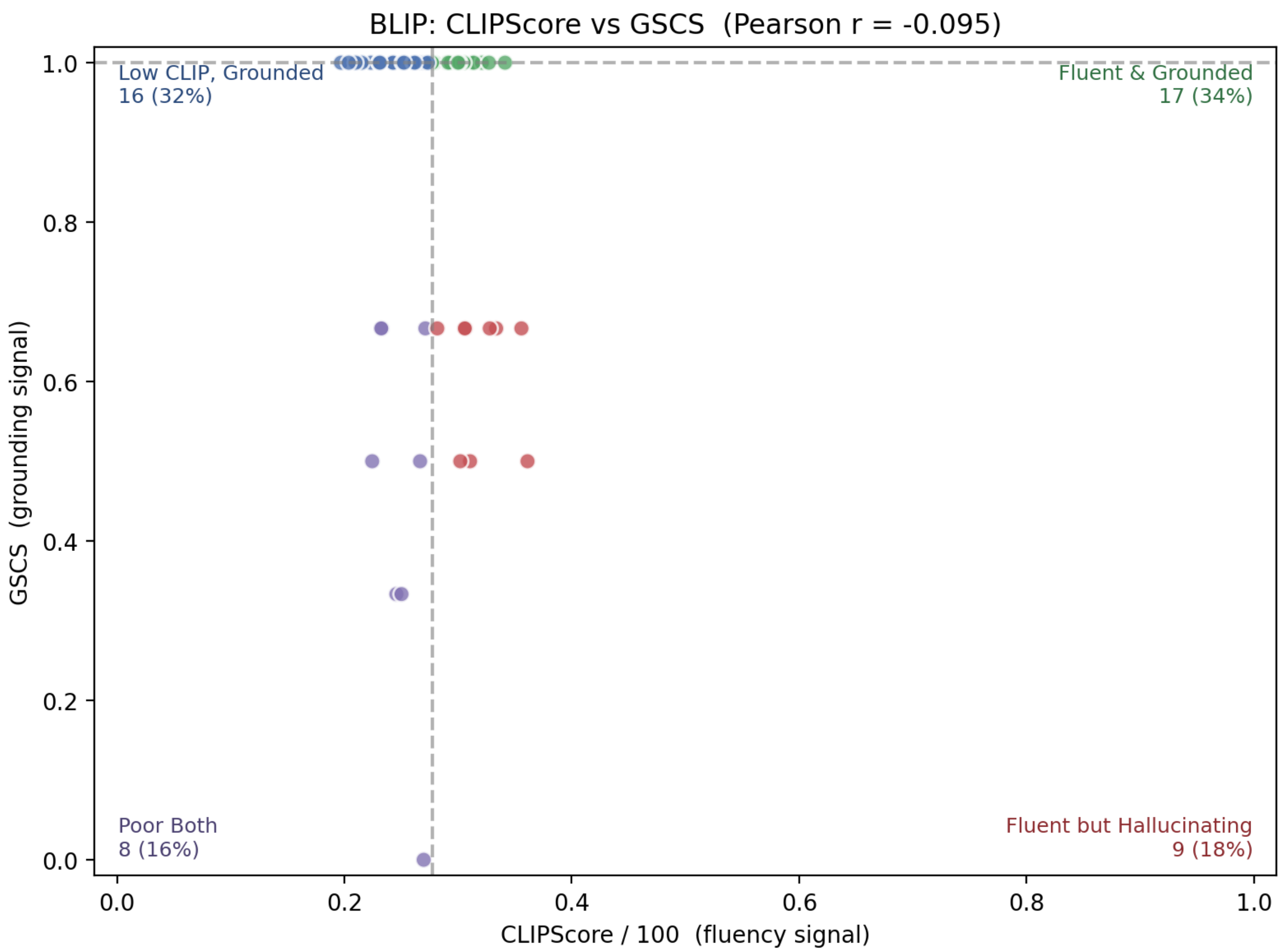

Figure 3. Per-image scatter of CLIPScore/100 (x) versus GSCS (y) for BLIP captions on the 50-image Flickr30k subset. The plot is partitioned into four quadrants by the per-axis median, and the title reports the Pearson correlation between the two signals. The bottom-right quadrant (high CLIPScore, low GSCS) contains captions that CLIP rates as fluent and on-topic but that mention objects that Grounding DINO cannot find. This is the central failure mode that motivates GSCS, since reference-based metrics and CLIPScore alone do not flag these captions.

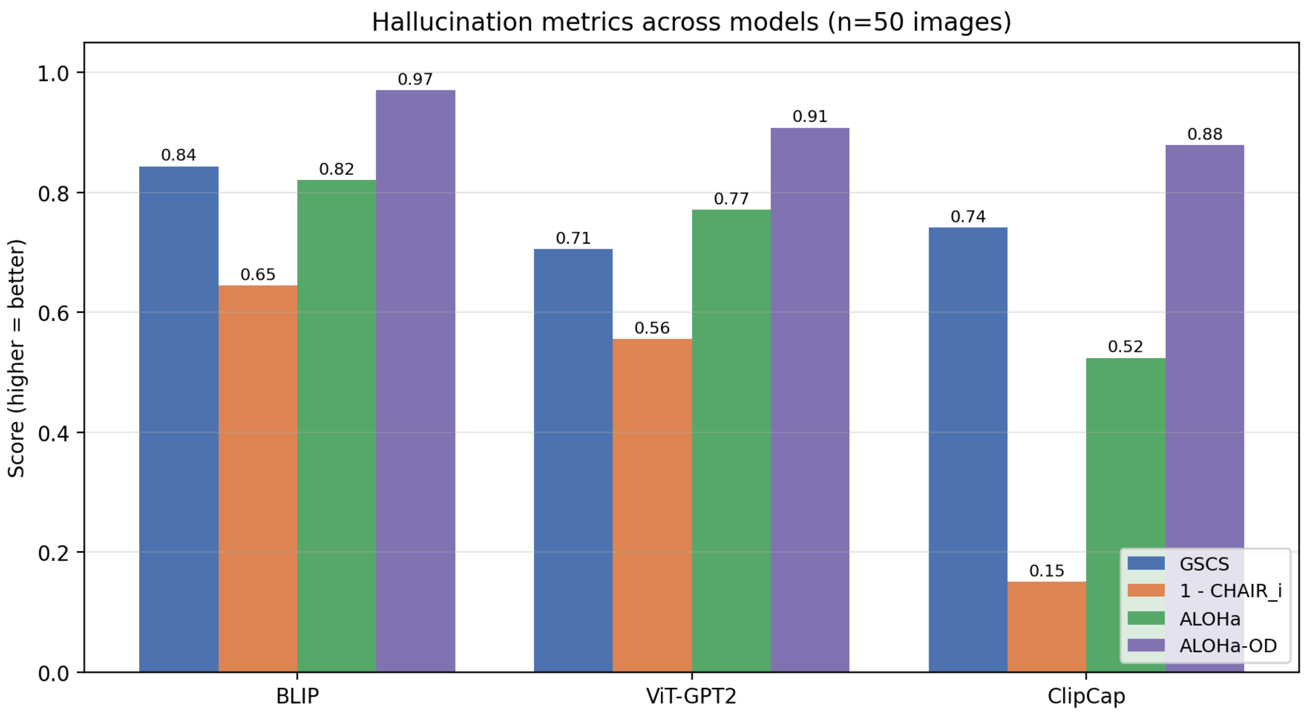

Figure 4. Four hallucination grounding metrics (GSCS, $1-$CHAIR_i, ALOHa, ALOHa-OD) per captioner on the same 50-image Flickr30k subset, all on a shared $[0, 1]$ axis where higher is better. BLIP wins under every metric. ClipCap is ranked second under GSCS but third under each of the three reference-based metrics, isolating the ranking swap discussed in §4.3 as a difference in evidence source rather than scoring rule.

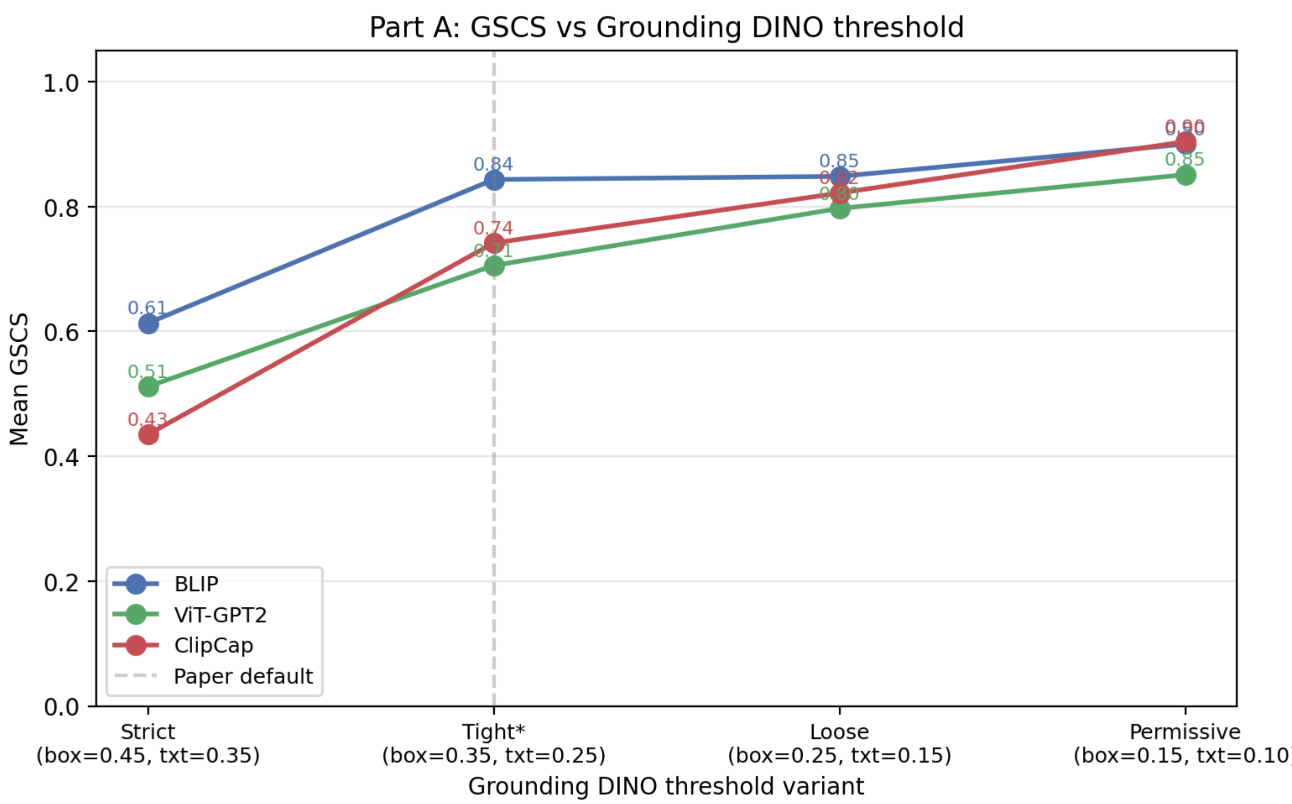

Figure 5. Mean GSCS for each captioner across the four Grounding DINO threshold variants from Table 2 (Strict, Tight* as the paper default, Loose, Permissive). All three lines rise monotonically as thresholds loosen, because lower confidence requirements admit more candidate detections. BLIP and ClipCap intersect near Permissive, which is the visual source of the threshold-induced ordering caveat in §4.2.1.